

labsheet=03 PRIYANSHU SHARMA CU240250980 SECTION(B) (47) Mastery in Data Analytics and Visualizations and Career Advancement (Technical)

```
# Statistical Analysis and Hypothesis Testing using Python
# Dataset: Titanic Dataset
# =====

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score
```

```
# 1. LOAD DATASET
# =====

print("\n" + "=" * 60)
print("1. LOADING TITANIC DATASET")
print("=" * 60)

try:
    df = sns.load_dataset("titanic")
except Exception:
    print("Unable to download Titanic dataset.")
    exit()

print("\nFirst 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())
```

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1. LOADING TITANIC DATASET
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First 5 rows:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	\
0	0	3	male	22.0	1	0	7.2500	S	Third	
1	1	1	female	38.0	1	0	71.2833	C	First	
2	1	3	female	26.0	0	0	7.9250	S	Third	
3	1	1	female	35.0	1	0	53.1000	S	First	
4	0	3	male	35.0	0	0	8.0500	S	Third	

	who	adult_male	deck	embark_town	alive	alone
0	man	True	NaN	Southampton	no	False
1	woman	False	C	Cherbourg	yes	False
2	woman	False	NaN	Southampton	yes	True
3	woman	False	C	Southampton	yes	False
4	man	True	NaN	Southampton	no	True

Dataset Shape:
(891, 15)

Column Names:
['survived', 'pclass', 'sex', 'age', 'sibsp', 'parch', 'fare', 'embarked', 'class', 'who', 'adult_male', 'deck', 'embark_tow

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# =====
# 2. BASIC INFORMATION
# =====

print("\n" + "=" * 60)
print("2. BASIC INFORMATION")
print("=" * 60)

print("\nDataset Information:")
print(df.info())
```

```
print("\nMissing Values:")
print(df.isnull().sum())
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2. BASIC INFORMATION
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```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male   891 non-null    bool
11  deck        203 non-null    category
12  embark_town  889 non-null    object
13  alive       891 non-null    object
14  alone       891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

```
Missing Values:
survived    0
pclass      0
sex         0
age         177
sibsp       0
parch       0
fare        0
embarked    2
class       0
who         0
adult_male   0
deck        688
embark_town  2
alive       0
alone       0
dtype: int64
```

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# 3. DESCRIPTIVE STATISTICS
# =====
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print("\n" + "=" * 60)
print("3. DESCRIPTIVE STATISTICS")
print("=" * 60)

# Numerical columns
numeric_columns = ["age", "fare", "sibsp", "parch"]

print("\nMean:")
print(df[numeric_columns].mean())

print("\nMedian:")
print(df[numeric_columns].median())

print("\nMode:")
print(df[numeric_columns].mode().iloc[0])

print("\nVariance:")
print(df[numeric_columns].var())

print("\nStandard Deviation:")
print(df[numeric_columns].std())

print("\nComplete Descriptive Statistics:")
print(df[numeric_columns].describe())
```

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3. DESCRIPTIVE STATISTICS
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Mean:
age      29.699118
fare     32.204208
```

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sibsp    0.523008
parch    0.381594
dtype: float64

Median:
age      28.0000
fare     14.4542
sibsp    0.0000
parch    0.0000
dtype: float64

Mode:
age      24.00
fare     8.05
sibsp    0.00
parch    0.00
Name: 0, dtype: float64

Variance:
age      211.019125
fare     2469.436846
sibsp    1.216043
parch    0.649728
dtype: float64

Standard Deviation:
age      14.526497
fare     49.693429
sibsp    1.102743
parch    0.806057
dtype: float64

```

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Complete Descriptive Statistics:

```

	age	fare	sibsp	parch
count	714.000000	891.000000	891.000000	891.000000
mean	29.699118	32.204208	0.523008	0.381594
std	14.526497	49.693429	1.102743	0.806057
min	0.420000	0.000000	0.000000	0.000000
25%	20.125000	7.910400	0.000000	0.000000
50%	28.000000	14.454200	0.000000	0.000000
75%	38.000000	31.000000	1.000000	0.000000
max	80.000000	512.329200	8.000000	6.000000

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# 4. PEARSON CORRELATION
# =====

print("\n" + "=" * 60)
print("4. PEARSON CORRELATION ANALYSIS")
print("=" * 60)

correlation = df[numeric_columns].corr(method="pearson")

print("\nCorrelation Matrix:")
print(correlation)

# Correlation heatmap
plt.figure(figsize=(8, 6))
sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)

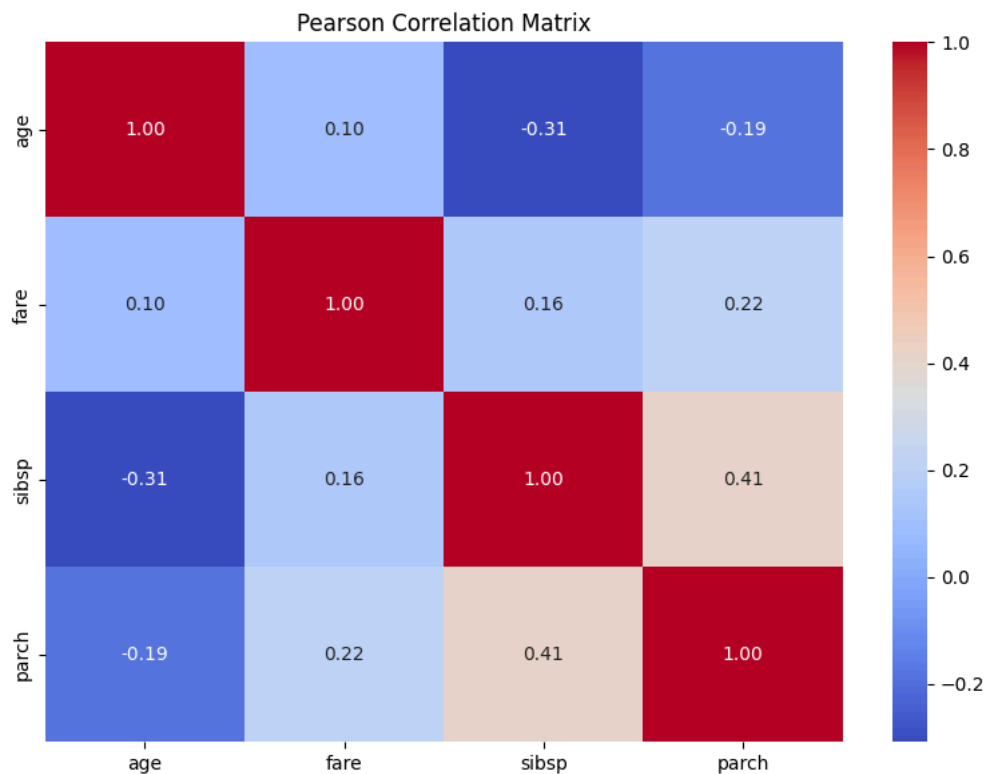
plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()

```

4. PEARSON CORRELATION ANALYSIS

Correlation Matrix:

	age	fare	sibsp	parch
age	1.000000	0.096067	-0.308247	-0.189119
fare	0.096067	1.000000	0.159651	0.216225
sibsp	-0.308247	0.159651	1.000000	0.414838
parch	-0.189119	0.216225	0.414838	1.000000



5. HYPOTHESIS TESTING

Independent Sample T-Test

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```
print("\n" + "=" * 60)
print("5. INDEPENDENT SAMPLE T-TEST")
print("=" * 60)

# Remove missing values
male_fare = df[df["sex"] == "male"]["fare"].dropna()
female_fare = df[df["sex"] == "female"]["fare"].dropna()

t_stat, p_value = stats.ttest_ind(
    male_fare,
    female_fare,
    equal_var=False
)

print("\nNull Hypothesis (H0):")
print("There is no significant difference between male and female average fare.")

print("\nAlternative Hypothesis (H1):")
print("There is a significant difference between male and female average fare.")

print("\nMale Average Fare:", male_fare.mean())
print("Female Average Fare:", female_fare.mean())

print("\nT-statistic:", t_stat)
print("P-value:", p_value)

alpha = 0.05

if p_value < alpha:
    print("\nResult: Reject H0.")
    print("There is a statistically significant difference.")
else:
    print("\nResult: Fail to Reject H0.")
    print("There is no statistically significant difference.")
```

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5. INDEPENDENT SAMPLE T-TEST
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Null Hypothesis (H0):
There is no significant difference between male and female average fare.

Alternative Hypothesis (H1):
There is a significant difference between male and female average fare.

Male Average Fare: 25.523893414211443
Female Average Fare: 44.47981783439491

T-statistic: -5.077499013458909
P-value: 5.388057409693902e-07

Result: Reject H0.
There is a statistically significant difference.
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# 6. ONE-WAY ANOVA
# =====

print("\n" + "=" * 60)
print("6. ONE-WAY ANOVA")
print("=" * 60)

# Compare fares among passenger classes

class_1 = df[df["pclass"] == 1]["fare"].dropna()
class_2 = df[df["pclass"] == 2]["fare"].dropna()
class_3 = df[df["pclass"] == 3]["fare"].dropna()

f_stat, anova_p = stats.f_oneway(
    class_1,
    class_2,
    class_3
)

print("\nNull Hypothesis (H0):")
print("There is no significant difference among the mean fares of the three classes.")

print("\nAlternative Hypothesis (H1):")
print("At least one class has a significantly different mean fare.")

print("\nClass 1 Mean Fare:", class_1.mean())
print("Class 2 Mean Fare:", class_2.mean())
print("Class 3 Mean Fare:", class_3.mean())

print("\nF-statistic:", f_stat)
print("P-value:", anova_p)

if anova_p < alpha:
    print("\nResult: Reject H0.")
    print("There is a statistically significant difference among groups.")
else:
    print("\nResult: Fail to Reject H0.")
    print("There is no statistically significant difference among groups.")
```

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=====
6. ONE-WAY ANOVA
=====

Null Hypothesis (H0):
There is no significant difference among the mean fares of the three classes.

Alternative Hypothesis (H1):
At least one class has a significantly different mean fare.

Class 1 Mean Fare: 84.1546875
Class 2 Mean Fare: 20.662183152173913
Class 3 Mean Fare: 13.675550101832993

F-statistic: 242.34415651744814
P-value: 1.0313763209141171e-84

Result: Reject H0.
There is a statistically significant difference among groups.
```

```
# 7. SIMPLE LINEAR REGRESSION
# =====

print("\n" + "=" * 60)
print("7. SIMPLE LINEAR REGRESSION")
print("=" * 60)

# Relationship between Age and Fare

regression_data = df[["age", "fare"]].dropna()

X = regression_data[["age"]]
y = regression_data["fare"]

model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

slope = model.coef_[0]
intercept = model.intercept_

r2 = r2_score(y, y_pred)

print("\nRegression Equation:")
print(f"Fare = {intercept:.2f} + ({slope:.2f} × Age)")

print("\nRegression Coefficient:", slope)
print("Intercept:", intercept)
print("R² Score:", r2)
```

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7. SIMPLE LINEAR REGRESSION
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Regression Equation:
Fare = 24.30 + (0.35 × Age)

Regression Coefficient: 0.34996368174402326
Intercept: 24.300901449288645
R² Score: 0.009228809267447624
```

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# 8. REGRESSION PLOT
# =====

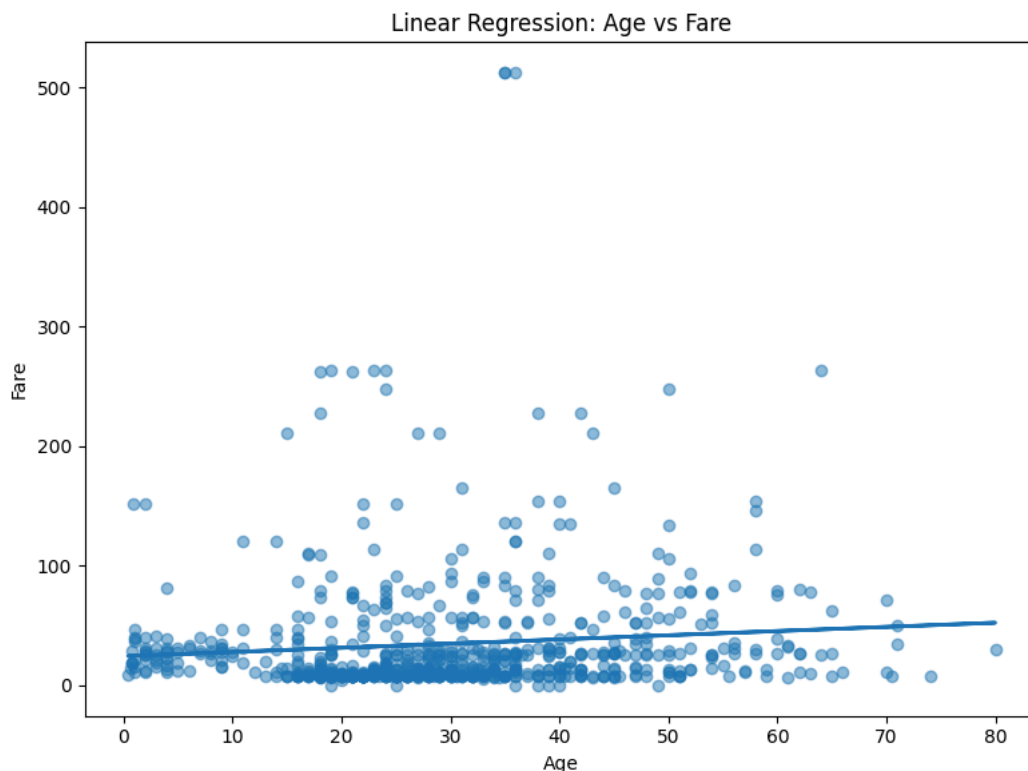
plt.figure(figsize=(8, 6))

plt.scatter(
    regression_data["age"],
    regression_data["fare"],
    alpha=0.5
)

plt.plot(
    regression_data["age"],
    y_pred,
    linewidth=2
)

plt.xlabel("Age")
plt.ylabel("Fare")
plt.title("Linear Regression: Age vs Fare")

plt.tight_layout()
plt.show()
```



```
# 9. CONFIDENCE INTERVAL
# =====

print("\n" + "=" * 60)
print("9. CONFIDENCE INTERVAL")
print("=" * 60)

# OLS regression for confidence interval

X_stats = sm.add_constant(X)

ols_model = sm.OLS(y, X_stats).fit()

print("\nOLS Regression Summary:")
print(ols_model.summary())

print("\n95% Confidence Intervals:")
print(ols_model.conf_int())
```

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9. CONFIDENCE INTERVAL
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OLS Regression Summary:

                        OLS Regression Results
=====
Dep. Variable:          fare      R-squared:                0.009
Model:                  OLS      Adj. R-squared:             0.008
Method:                 Least Squares      F-statistic:         6.632
Date:                   Mon, 24 Aug 2026    Prob (F-statistic):    0.0102
Time:                   08:23:08           Log-Likelihood:       -3843.0
No. Observations:       714              AIC:                7690.
Df Residuals:           712              BIC:                7699.
Df Model:                1
Covariance Type:        nonrobust
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	coef	std err	t	P> t	[0.025	0.975]
const	24.3009	4.492	5.410	0.000	15.481	33.120
age	0.3500	0.136	2.575	0.010	0.083	0.617

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=====
Omnibus:                 716.815      Durbin-Watson:          1.925
Prob(Omnibus):            0.000      Jarque-Bera (JB):       30883.661
Skew:                     4.654      Prob(JB):               0.00
Kurtosis:                 33.846      Cond. No.:              75.3
=====
```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

95% Confidence Intervals:

	0	1
const	15.481351	33.120452
age	0.083165	0.616762

```
# 10. FINAL SUMMARY
# =====

print("\n" + "=" * 60)
print("10. FINAL SUMMARY")
print("=" * 60)

print("""
Statistical analysis was performed on the Titanic dataset.

1. Descriptive statistics were calculated using mean, median,
   mode, variance and standard deviation.

2. Pearson correlation was used to find relationships between
   numerical variables.

3. Independent Sample T-Test was used to compare the average
   fare between male and female passengers.

4. One-Way ANOVA was used to compare fares among passenger
   classes.

5. Linear Regression was used to study the relationship between
   age and fare.

6. P-values were used to determine statistical significance.

7. R2 score was used to measure how well the regression model
   explains the variation in the dependent variable.

Statistical analysis helps in making data-driven decisions and
understanding relationships between variables.
""")

# =====
```

```
=====
10. FINAL SUMMARY
=====

Statistical analysis was performed on the Titanic dataset.

1. Descriptive statistics were calculated using mean, median,
   mode, variance and standard deviation.

2. Pearson correlation was used to find relationships between
   numerical variables.

3. Independent Sample T-Test was used to compare the average
   fare between male and female passengers.

4. One-Way ANOVA was used to compare fares among passenger
   classes.

5. Linear Regression was used to study the relationship between
   age and fare.

6. P-values were used to determine statistical significance.

7. R2 score was used to measure how well the regression model
   explains the variation in the dependent variable.
```